IoT Approach Application for Development of Autonomous Beekeeping System

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***Abstract*— In this paper we discuss the offered autonomous beekeeping system' prototype, which applies Internet of Things (IoT) and systems automation approach known as the Arrowhead framework. The proposed solution is based on the services provided by the Event Handler system. The proposed smart services broker applies a visual data flow programming approach. A Node-RED is used to prove viability and advantages of offered architecture implemented in a local automation cloud of autonomous beekeeping System of Systems. The autonomous beekeeping system' prototype provides useful data on beehive and bee apiary status (internal and ambient temperature, humidity, weight of the hives, etc.) to the users, so they can evaluate the hive status and take further action. Therefore, the use of autonomous beekeeping system provides opportunity to increase bee productivity and to save cost of maintenance of beekeeping.**

Keywords—Wireless sensor networks, Arrowhead framework, Internet of Things, Precision Beekeeping.

I. Introduction

Nowadays beekeeping is currently based largely on manual work that requires regular visits to bee apiaries and to monitor bee hives. However, physical inspection interferes bees' life and causes stress that negatively affects the productivity of all product lines. By implementing the autonomous monitoring system, the hives conditions can be tracked remotely, e.g. whether the inside temperature is critical, if the family is missing feed, therefore any critical deviation can be detected and prevented in time. Such system could be useful for remote monitoring of bee hive' profitability, flying activities, bee colony moods and bees' family health status. The notion of Precision Beekeeping (PB) is defined [1], [2] as an apiary management strategy based on monitoring of individual bee colonies to minimize the resource consumption and maximize the productivity of bees. There are a few examples of approaches and solutions: temperature and humidity controlling system, which is powered by solar energy [3]; open-source bee-hive monitoring system developed by Glyn and Clive Hudson [4]. An IoT system prototype for honey bee colony temperature monitoring is described in [5].

The project "Autonomous Beekeeping" was initiated by the Riga Technical University, Latvian Internet Association and two beekeeper farmers at the beginning of 2018. The project is funded by the European Agricultural Fund for Rural Development Program 2014-2020 Cooperation: Support new products, methods, processes and technologies. The part of the research: communication system, is implemented in a frame of COST Action CA15127 Resilient communication services protecting end user applications from disaster-based failures – RECODIS.

In this project we apply Internet of Things approach for development of monitoring and control system of bee colony activities. Such system is based on service oriented architecture (SOA) [7], [8], IoT approach and advanced industrial automation standards in order to provide highly compatibility with existing and emerging solutions. In this context the IoT has been recognized as a promising solution since it has proved its applicability in very diverse areas such as public utility systems automation [9], efficient transportation systems [10], centralized healthcare [11], efficient waste management, smart grids, digital tourism, etc.

In this sense a cloud service platform for IoT has been offered [12]. Derhamy et al. [13] categorized a number of cloud service platforms as global cloud, peer-to-peer and local cloud. Furthermore, often "fog" or hybrids of global and internal clouds are positioned in between the global and local cloud paradigms. The local cloud concept developed by the Arrowhead Framework [14] addresses many challenges related to IoT-based automation, and is unique in its support for integration of applications between secure localized clouds. The approach is that IoT devices are abstracted as services in order to enable interoperability between IoT devices [15]. A macro-programming model capable to oversee the network of distributed sensors as a whole rather than to program individual pieces is suggested by [16]. Giang [17] and Blackstok et al. [18] offered an advanced version of dataflow model to express application logic of IoT devices suitable for large scale IoT, called as "Adaptive Distributed Dataflow".

During a summer season beekeepers usually reallocate bee apiaries several times pursuing better conditions for harvesting of honey products. This fact sets requirement to the autonomous beekeeping system: a wireless network, which ensures data collection from sensors, has to be easy and fast setup, independent from electricity supply, reliable, safe and independent of weather conditions (rain, fog, froze, etc.).

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The offered solution will perform bee apiary control (individual bee colonies and at the apex level) without interfering with its processes, while optimizing frequency of the apiary inspection. The system will help to analyze data correlation with video, meteostation data, mass changes in time as well as interpretation of nest temperature, humidity and linking to local geographic and biological conditions.

In this paper, we discuss IoT approach for autonomous beekeeping, based on SOA when sensors data exchange is made through the services. The proposed technology is based on the Arrowhead Framework [19]. The proposed solution is based on the services provided by the Event Handler system. The proposed smart services broker applies a visual data flow programming approach [18]. A Node-RED [20] was selected among other tools to prove viability and advantages of offered architecture implemented in a local automation cloud of autonomous beekeeping System of Systems. The services broker resides on a gateway, which provides adapter and protocols translation functions, and on a tool for wiring together hardware devices, and APIs for online services.

The paper is structured as follows. A notion of the Precision Beekeeping and overview of related works is done in Introduction. In the Chapter 1 authors give a brief description of the architecture of Autonomous beekeeping system, which applies the Arrowhead Framework. Implementation of Autonomous beekeeping system is described in the Chapter 2. The review of main contribution of the research and possible directions for future works are discussed in the Chapter 3.

II. Architecture of Autonomous Beekeeping System

A. *Arrowhead Framework Approach for the Autonomous beekeeping*

For creation of Autonomous beekeeping system (AB system), we applied the Arrowhead Framework approach. The Arrowhead Framework (AF) includes a set of Core Services, which support the interaction between Application Services (e.g. sensor readings, controlling of actuating devices, energy consumption, temperature measurement services, etc.). The Core Services handle the support functionality within the AF to enable application services to exchange information. The AF is built upon the local cloud concept, where local automation tasks should be encapsulated and protected from outside interference. Each local cloud must contain at least the three mandatory core systems: Discovery, Authorization and Orchestration, which allow to establish connections between Arrowhead application services [19].

The core system, named as Event Handler System [21], provides functionality for the handling of events that occur in an Arrowhead network. The Event Handler (EH) receives events from Event Producers and dispatches them to registered Event Consumers. The EH logs events to persistent storage, registers producers and consumers of events and applies filtering rules to events distribution.

In our research we developed Event Handler System as a service broker, which enables SOA based services and data flow between diverse type of embedded devices and nodes, for example: humidity, wind strength, weight, outdoors/ indoor temperature sensors, etc. We implemented Event Handler core system as a MQTT (Message Queuing Telemetry Transport) service broker that applies a software integration platform Node-RED [20] to interconnect heterogeneous systems in IoT way. MQTT is an ISO standard (ISO/IEC PRF 20922) publish-subscribe-based messaging protocol. It works on top of the TCP/IP protocol. It is designed for connections with remote locations where the network bandwidth is limited. A MQTT system consists of clients communicating with a server, usually called a "broker". More detailed description is provided in the Chapter 3.

B. *The system model*

In order to create a model of Autonomous Beekeeping (AB) system we use a methodology represented by OMG's (Object Management Group) solution for system abstraction, modelling, development, and reuse—Model Driven Architecture (MDA) [6]. The key component of a system modelling, which underlies the principles of MDA—Unified Modelling Language (UML)—is a widely accepted standard for modelling and designing different types of systems. In order to create the UML model of AB system we apply Enterprise Architect Professional modelling tool.

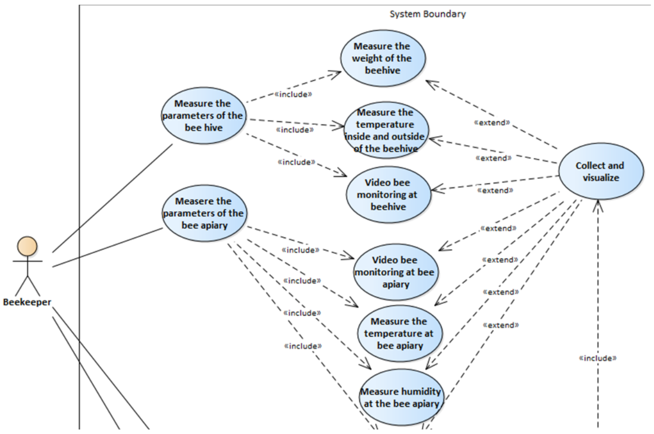

Fig. 1. Use case diagram of AB system

Fifteen primary use cases have been defined aiming to describe AB model (see Fig.1). Among them are Collect and visualize parameters, Measure beehive weight, Measure beehive's internal temperature, Measure the beehive's internal moisture, Monitor energy consumption, etc.

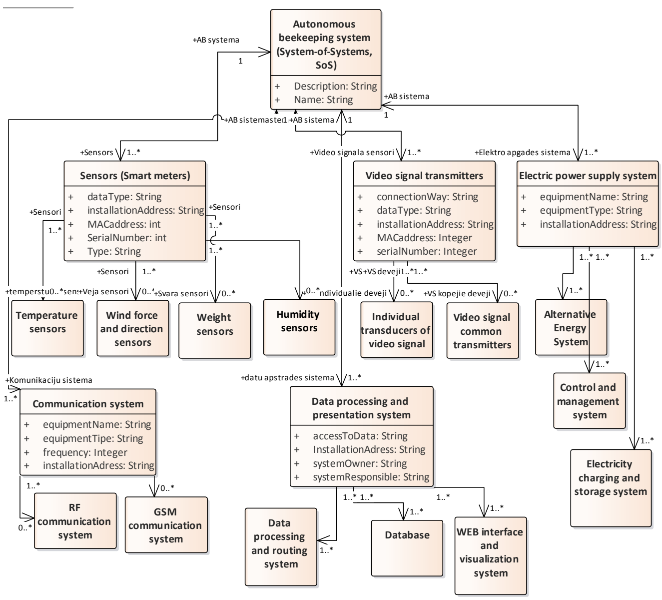

Fig. 2. Domain Objects diagram of AB system

We use the actor symbol to represent the agent that activates the use case, probably the beekeeper or the other AB system' client, which is willing to analyze the behavior of a bee colony.

A Domain Objects diagram at Fig. 2 shows nineteen classes, where four of them represent different kinds of sensors with its services components, four classes represent power supply and energy storage, four classes data storage processing and visualization services, two classes data communication services, and etc. The parameters of attributes and operations in classes have been omitted in the interest of figure clarity.

III. System functionality and composition

In this research, we tested particular elements of the Autonomous Beekeeping system installed at five of bee hives in a realy field conditions of the Riga Botanic Garden. During testing, weight of bee hives, humidity as well as the temperature sensors, which measure the temperature inside of the beehives and outside temperature, were tested (see Fig.3).

A. Communication System

In this research, we developed Wireless Sensor Networks (WSN) that are widely used for implementation of Internet of Things (IoT) concept at the different areas of application.

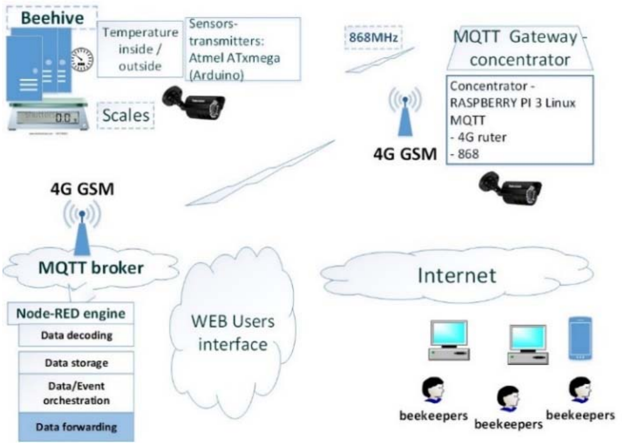

Fig. 3. A bock-sheme of beehive monitoring system.

A communication system of AB System-of-Systems comprise sensors-transmitters and gateways. The sensors-transmitter consists of a microcontroller, readout interface, power supply and ISM band radio module operating at 868MHz. The transmission range of sensors-transmitters is up to several hundred meters depending on the installation environment. The measurement data are encapsulated in telegrams to be transmitted using Manchester coding and GFSK modulation to one or to multiple gateways that provide an area of coverage.

The gateway node consists or a radio module, GSM GPRS module and /or 802.11b/g/n Wi-Fi module that integrates a microcontroller (ESP8266). The Wi-Fi module supports both station: Wi-Fi client mode and access point modes. On startup the gateway tries to connect to a Wi-Fi access point to access internet, if any of access points is not configured or not reachable, the gateway enters access point mode and starts broadcasting a SSID (Service Set Identifier) named after its MAC address. The owner of the gateway or the operator can use any PC or mobile equipment supporting 802.11b/g/n to connect to the gateway using a pre-shared password. The gateway automatically redirects a connecting client device to a configuration web interface. Particularly for AB system a 4G reliable and secure LTE router with I/O, GNSS and RS232/RS485 for professional applications was used for communication between gateway-concentrator and MQTT broker at the back-end. Router delivers mission-critical cellular communication and GPS location capabilities.

The sustainability issues of autonomous beekeeping have been considered in [22], [23] with the research focus on bees behavior in different environment condition. The research [24] investigated a metering solution based on the modular principle for inter-system communication paradigm using selectable interface modules (IEEE 802.3, ISM radio interface, GSM/GPRS).

In our research we put emphasis on the development of a reliable monitoring system, which should be survivable in changing outdoor weather conditions: significant deviations of the temperature, humidity, rain, froze, snow, frosting impact, etc. Therefore, data transmitting in different weather conditions have been tested in order to evaluate resilience of AB system against variable weather conditions.

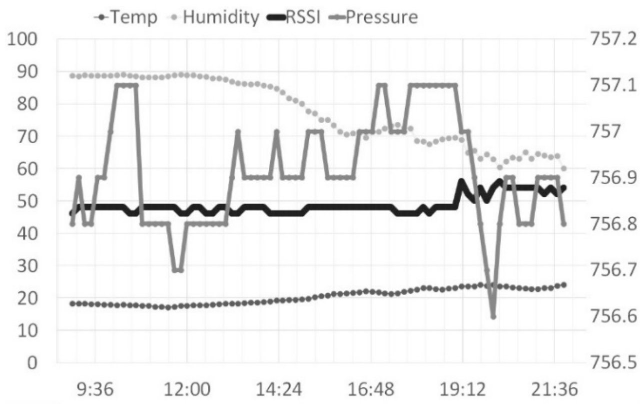

Fig. 4. Environment factors and RSSI Correlation (the values at the left vertical axis are in percent, right vertical axis are mm Hg (torr), horizontal axis are in hours: minutes).

The validation of reliability of data transmission can be implemented by alerting a service performance degradation. The weather impact on the tested system was assessed by comparison RSSI (Received Signal Strength Indicator) data and the weather conditions data from the Weather monitoring system of Botanical garden in Riga. It was noted that changes of humidity level caused the strength of the signal more than other factors, however the RSSI deviations were within the acceptable limit (see Fig.4).

The weather impact, measured during summer period, was used as a reference for evaluation of Quality of Delivery (QoD). The actual topology of WSN is defined based on the energy consumption level of nodes, distances between bee hives location points, and weather conditions at defined geographical locations in order to assure reacquired level of Quality of Delivery (QoD). Due to RSSI deviation happened within the acceptable range, we could conclude that the tested AB system' prototype showed fair sustainability in different weather conditions.

B. Video monitoring system

A video monitoring system is created and tested in addition to sensor data collected from beehives. The purpose of testing is to test the individual elements of the Autonomous

Beekeeping system under practical conditions, that is, one of bee apiaries. During the testing, it was planned to install video surveillance camera and video recording equipment on bee hives and perform video surveillance capabilities. For this purpose, two cameras were installed (Fig.5):

- A general review camera that allows you to monitor all hives.
- An individual camera at one of the hives to observe the movement of the bees from and to the hive.

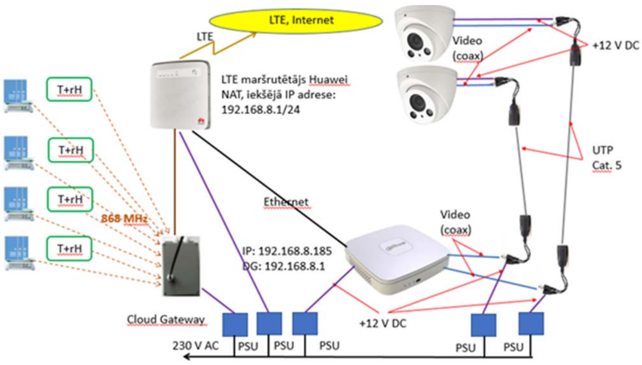

Fig. 5. Video monitoring block-scheme

The equipment for test system was used:

- 2x outdoor AHD cameras Dahua.
- DVR Dahua DVR-4104.
- LTE router for Huawei E5186s.
- Cloud Gateway E-Meter.

In order to provide access to the Internet, the operator Bite service was used, which includes necessary equipment - the Huawei LTE router. The DVR and the Cloud Gateway equipment were connected to the Ethernet ports of the Huawei router. The router provides DHCP dynamic address assignment, but for ease of administration, the DVR was set to a static address. The Huawei router also provided Wi-Fi connections, which is used for on-site work, namely, to receive video streams and to regulate, administer and configure the DVR.

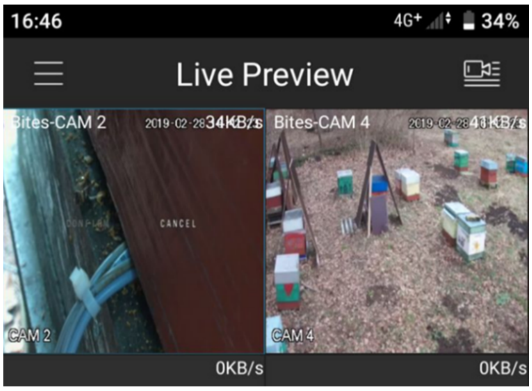

Fig. 6. Video monitoring: two views – general view and individual at particular beehive

Mobile Internet LTE service provides Internet access via Network Address Translation (NAT) and Port Address Translation (PAT) via mobile operator firewall. An internal (non-routable) IP addresses are assigned to the LTE router external interface. The video surveillance provides two views (see Fig. 6).

C. Data processing and interpretation

The beehive monitoring system deployed a Node-RED [20] as a gateway concentrator for wiring together hardware devices (temperature, weight, humidity sensors, etc.), APIs and online services. At the heart of Node-RED is a visual editor allowing complex data flows to be wired together with a little coding skills. Node-RED main functionality is to decode and to route MQTT smart metering data to further service orchestration or use them in external services as monitoring system or external clients (see Fig. 3 and 8). The measurements from the sensor-transmitters are transmitted to the concentrator from sensors using ISM (Industrial, Scientific and Medical) range signal 868 MHz (for example, weight sensor 1C3003AE0030 transmitted data via radio gateway-concentrator).

Fig. 7. Outdoor weight measurement installation.

The temperature sensor-transmitters transfer data to the server-broker every 15 minutes, the weight sensor transmits data to the server every 5 minutes. The data collected by the MQTT broker are displayed in graphical form. The users interpret data in graphics or as "dashboard", based on their own experience and understanding of ongoing processes in the hive.

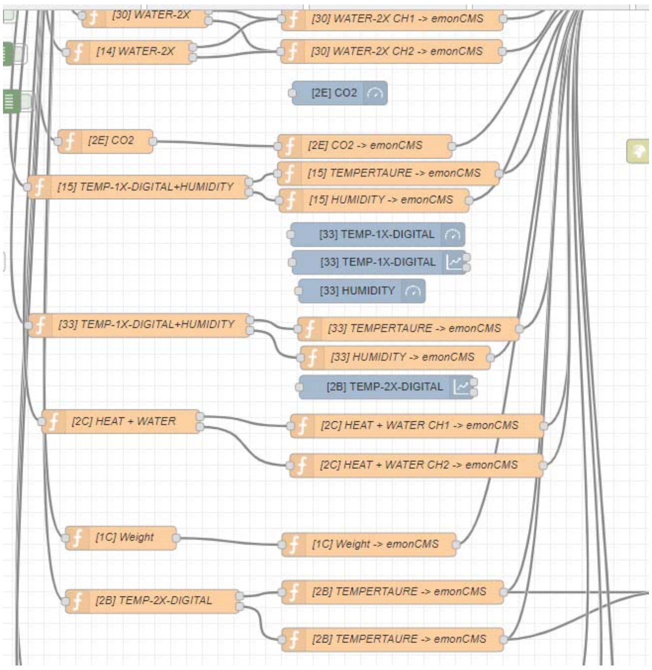

Fig. 8. The nodeRED workflow for wireing of different devices and data processing.

For monitoring of the temperature two sensors are allocated inside of a beehive: one in the center and the second one in the side part of a beehive protected from sunlight. The third sensor is allocated at the outside wall of the beehive to get measurements of ambient air temperature. The weight of each hive is measured by a specially designed weight platform (Fig. 7), and measurement data are sent by a sensor-transmitter to the gateway-concentrator without additional processing.

Node-RED main functionality is to decode and to route MQTT smart metering data (Fig. 8) to further service orchestration or for use in external services as customer billing or monitoring systems. An Inject node is a testing element to generate a simulated payload for debugging purposes. Below is a sample payload for a combined sensor with weight data:

MQTT payload: `00:04:40 42 0B 1C3003AE0030 0337 002580`, where 1C3003AE0030 is a sensor ID, RSSI LENSN03, 37 (battery level) and 002580 is data read out.

The task is to decode the payload and forward it to a data storage and visualization service using the IoT approach. The first step is to define a MQTT source, which is a gateway. Gateways post received metering data to topic based on their MAC address that also serves as a configuration service by subscribing to MAC configuration. The node "SERIAL+VALUE" creates an array of elements from the initial payload (telegram); each array element is the processed separately depending on the needed of output or post-processing. A new object is created by dividing name/value pairs, where the crucial elements are: serial number – that identifies the data type and coding format and the unique device id, which makes the metering devices distinct. The second value is the data block needed for decoding procedures. Double output creates two new MQTT messages containing separated data from each sensor, with measurement type topic and devices unique serial number (node Weight).

For further processing (e.g. monitoring, for serving of external clients) emoncms [25] code base is used (node Wight->emonCMS). From the example above a type+device_id object name is generated. After data delivery to emoncms, these data objects are discovered as inputs. Inputs are generic variable that can be processed using emoncms processing engine.

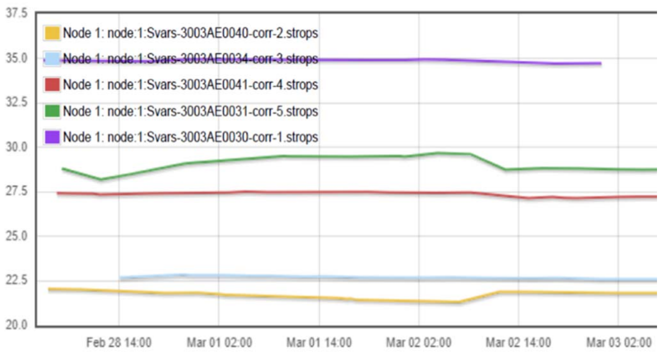

Fig. 9. Web based interface of on-line published monitoring results. Measurements of the weight deviations.

Emoncms represents measurement data as a dashboard and charts. For example, the print screen shows the deviations of five beehives weight (see Fig.9).

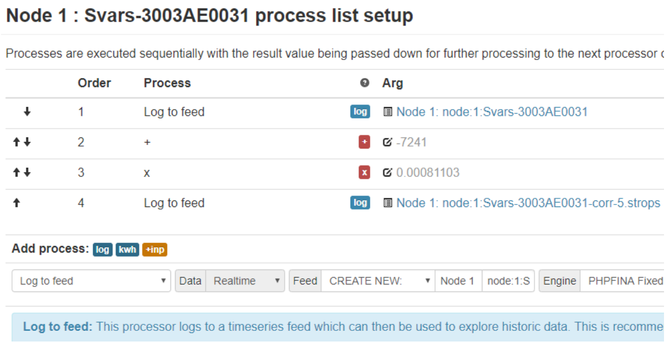

Fig. 10. An algorithm of the block for Query Emoncms API

Before saving information for analysis and representation, pre-calculation of sensors measurement offset values should be done in order to calculate the actual sensor readout value. At Fig.10 an example of "log to feed" value pre-calculation for a hive weight is depicted, when pre-calculations are done in four steps. Users have to customize data transformation, delta offsets, calibration, scaling etc. Typically, the broker does not save MQTT data. This can be done by defining a flow to a data storage, like MySQL DB. In this application, data storage is defined in emoncms with the "Log to feed" processor.

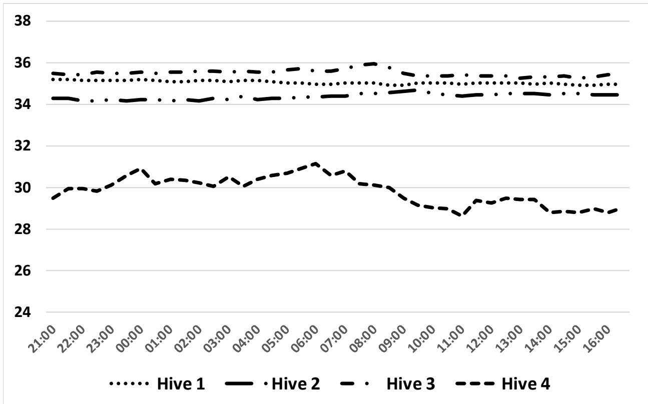

Fig. 11. Multi-chart view of temperature measurement inside of 4 beehives

Beekeepers and the other project partners are interested to compare behavior of bee colonies and particularly, at the transitions of the seasons, when bees start awakening to leave hives after the winter time. A new version of AB provides opportunity of comparison of bee colonies thanks to multi-chart view (see Fig.11).

IV. Conclusions and Next Steps

The offered autonomous beekeeping system' prototype applies IoT and systems automation approach known as the Arrowhead framework. AB system' prototype provides useful data on hives and bee apiary status (internal and ambient temperature, humidity, weight of the hives, etc.) to the users, so they can evaluate the hive status and take further action. Therefore, the use of autonomous beekeeping system provides opportunity to increases bee productivity and to save cost of maintenance of beekeeping.

Development of autonomous beekeeping system is still in progress, since the project team implemented the first stage of the project. Further research should be focused to the crucial issues in order to make AB system really "autonomous". First of all, it is necessary to ensure full autonomous energy supply. Therefore, it is necessary to evaluate consumption of energy by gateways-concentrators and routers against the opportunity to apply alternative energy sources, such wind and photovoltaic elements.

The evaluation of the weather impact at the deviation of the parameters, which characterize QoS (Quality of Service) of WSN [26], was done in the summer season, therefore more deep assessment of OoS of the AB system should be done in the real harsh weather conditions (e.g. in the winter time).

Additionally, it is planned to replicate the prototype of AB system at the second bee apiary at a different location taking into account experience obtained at the first project stage.

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